Composition and Inference for Covariate-Indexed Parameter Paths in Recursive Conditional Models

Decision

A unified theory exists, but its useful theorem is a certificate calculus rather than one universal persistence inequality. The common object is a finite-dimensional, charted regression for a covariate-indexed predictable parameter path, composed with a recursive conditional model. A nonnegative matrix records how each state block reacts to every other block. Contraction of products of these matrices gives existence and uniform filter forgetting for the joint state. A separate Banach-space theorem gives stable first- and second-derivative recursions. Standard likelihood arguments then deliver estimation and inference, but only after the composition and derivative certificates have done the model-specific work.

The ordinary fixed model is the exact zero-slope restriction. This supports a regular Wald or score test when the slope directions remain identified and interior under the null. It does not make boundary cases, unidentified nuisance parameters, score-driven modifiers, nonparametric curves, or arbitrary scoring rules regular. Those cases need separate theorems. The framework is therefore common in representation, composition, differentiation, and regular inference. Parameter domains, contraction, moments, nonsmooth points, and nonregular limits remain model-owned.

1 What is established, and what remains

Varying-coefficient regression supplies the function representation [23]; functional-coefficient autoregression places it inside a time-series equation [10]. General causal-process theory supplies stationary quasi-likelihood results for broad fixed-parameter and exogenous-covariate models [3, 15]. Smooth deterministic variation has its own locally stationary theory, including recursive ARMA and GARCH examples and uniform inference [12, 4, 21, 42]. These results already cover much of the probability and estimation theory for their declared classes.

Random Lipschitz maps supply the state mechanism [9, 13]. The same mechanism underlies likelihood theory for conditional heteroskedasticity and observation-driven models [34, 16]. Score-driven models already have primitive conditions for identification, invertibility, consistency, and asymptotic normality [38]. Feasible empirical invertibility regions are also established for observation-driven filters [37]. The contribution below is therefore not generic observation-driven asymptotics or contraction itself.

The narrower gap is compositional. Existing work does not give a component-level calculation that: (i) combines cross-block state Lipschitz bounds; (ii) propagates the first two coefficient derivatives through the joint recursion; (iii) pulls a charted path likelihood back to one coefficient vector; and (iv) connects that calculation to regular fixed-path tests. The results below fill that gap for covariate-indexed paths whose modifiers are fixed across candidate likelihood evaluations. Candidate-dependent score recursions enter only as additional state blocks.

2 The parameter-path recursive conditional model

2.1 Timing and information

Let $Y_t \in \mathcal{Y} \subseteq \mathbb{R}^m$ be the observation at date t. Let $\mathcal{F}_t = \sigma(Y_s, X_s : s \le t)$ be the post-observation information set. Let C_t contain the information available after the date-t modifiers are known but before Y_t is observed. The timing convention is

$$\mathcal{F}_{t-1} \subseteq C_t \subseteq \mathcal{F}_t, \qquad \mathcal{F}_t = C_t \vee \sigma(Y_t), \qquad X_t \in C_t, \qquad Y_t \notin C_t, \tag{1}$$

where $X_t \in \mathcal{X}$ is the modifier. Equation (1) permits predetermined date-t information and lagged observations. It excludes contemporaneous or future information unless the model specifies their joint law. Predictability alone does not make X_t external. If X_t depends on past observations, the existence proof must cover the joint recursion. The realized path $X_{1:n}$ is held fixed when the criterion compares candidate values c. A candidate-dependent filtered modifier belongs in the recursive state, not in X_t.

2.2 Basis, chart, and parameter path

Partition the model parameters into G blocks. Block g has an unconstrained coordinate of dimension r_g, a basis vector $b_g(X_t) \in \mathbb{R}^{k_g}$, and a coefficient matrix $C_g \in \mathbb{R}^{r_g \times k_g}$. Define

$$b_g(X_t) = \begin{pmatrix} 1 \\ \widetilde{b}_g(X_t) \end{pmatrix}, \tag{2}$$

$$\eta_{g,t}(C_g) = C_g b_g(X_t), \tag{3}$$

$$\vartheta_{g,t}(C_g) = T_g\{\eta_{g,t}(C_g)\} \in \Theta_g^\circ. \tag{4}$$

Here $\widetilde{b}_g$ contains the nonconstant basis functions, $T_g : U_g \to \Theta_g^\circ$ is a model-owned map, and Θ_g° is the interior of the block's theoretical parameter domain. The represented path must satisfy

$$\Pr\left[\eta_{g,t}(c) \in U_g \text{ for every } c \in \mathcal{B}\right] = 1. \tag{5}$$

Injectivity is enough for exact nesting and path uniqueness. Regular first-order inference also requires a full-rank differential near the true path. A block must contain every parameter coupled by a restriction. Examples include an AR root vector, a GARCH persistence allocation, a correlation matrix, and a jointly standardized density-shape vector.

For an affine specification with a scalar modifier x_t, equation (2) becomes

$$b_g(x_t) = (1, x_t)', \qquad \eta_{g,t} = a_g + d_g x_t, \tag{6}$$

where $a_g \in \mathbb{R}^{r_g}$ is the baseline coordinate and $d_g \in \mathbb{R}^{r_g}$ is the slope. Linear variation is therefore linear in the declared chart. An affine log-variance path and an affine variance path are different statistical models.

The nonconstant basis needs a normalization. A population-centering convention is

$$\mathbb{E}\{\widetilde{b}_g(X_t)\} = 0, \tag{7}$$

while a reference-value convention is

$$\widetilde{b}_g(x_g^\star) = 0. \tag{8}$$

Equation (7) makes a_g a baseline at the mean modifier on the link scale. Equation (8) makes it the link-scale value at $x_g^\star$. Neither convention makes $T_g(a_g)$ equal to the mean raw parameter when T_g is nonlinear.

Let $c \in \mathcal{B} \subseteq \mathbb{R}^p$ stack $\mathrm{vec}(C_1), \ldots, \mathrm{vec}(C_G)$. Let $\vartheta_t(c)$ stack the block paths from equation (4). The map $c \mapsto \vartheta_{1:n}(c)$ is the parameter-path map.

2.3 Recursive state and conditional law

Let (S, d) be a complete separable metric state space. The state may contain observation lags, innovation lags, conditional variances, log variances, powered scales, density states, and positive-definite dependence matrices. Define the observed-data state recursion by

$$S_t^c = \Psi\big(S_{t-1}^c, Y_{t-1:t-L}, X_{t:t-M}, \vartheta_{t:t-M}(c)\big), \qquad S_t^c \in \mathsf{S}, \tag{9}$$

where L and M are finite history lengths after any required state augmentation. Infinite-memory models use a state in an appropriate sequence space or an equivalent recursive representation.

Let μ be a common dominating measure on the observation space. The conditional law is

$$Y_t \mid C_t \sim p_c(\cdot \mid S_t^c, X_t)\, \mu(dy). \tag{10}$$

Equation (10) defines a likelihood conditional on the modifier process. It is a joint likelihood for (Y_t, X_t) only if the model also specifies the conditional law of X_t. If the same parameter governs both laws, omitting the modifier law changes the objective.

The exact stationary infinite-past negative log likelihood on the scored sample $\mathcal{I}_n = \{t_0, \ldots, n\}$ is

$$Q_n(c) = \sum_{t \in \mathcal{I}_n} q_t(c), \qquad q_t(c) = -\log p_c(Y_t \mid S_t^c, X_t), \qquad n_{\mathrm{eff}} = |\mathcal{I}_n|. \tag{11}$$

The numerical filter starts from a declared seed and produces

$$\widetilde{Q}_n(c) = \sum_{t \in \mathcal{I}_n} \widetilde{q}_t(c). \tag{12}$$

The estimator minimizes equation (12); equation (11) supplies the stationary comparison object. The first scored date, histories, and conditioning seeds form the sample plan. Dividing equation (11) by n_{eff} may help an optimizer, but it does not redefine the likelihood.

We call equations (1)–(12) a *covariate-indexed predictable parameter-path recursive conditional model*. The name identifies the three mathematical objects that matter: a predictable parameter path, a recursive state, and a conditional law.

3 The constant model is an exact restriction

Write each coefficient matrix as

$$C_g = \begin{bmatrix} a_g & A_g \end{bmatrix}, \tag{13}$$

where $A_g \in \mathbb{R}^{r_g \times (k_g - 1)}$ contains every nonconstant coefficient. Define the fixed subspace by

$$\mathcal{B}_{\mathrm{fix}} = \big\{ c \in \mathcal{B} : A_g = 0 \text{ for } g = 1, \ldots, G \big\}. \tag{14}$$

Proposition 1 (Exact fixed-model nesting)**.** *Suppose T_g is onto the interior domain used by the constant model. For each $c \in \mathcal{B}_{\mathrm{fix}}$, the parameter path, state recursion, conditional density, likelihood value, and intercept-block derivatives equal those of the constant-parameter model at $\bar{\vartheta}_g = T_g(a_g)$, provided both models use the same sample plan.*

For $c \in \mathcal{B}_{\text{fix}}$, equation (4) gives

$$\vartheta_{g,t}(c) = T_g(a_g) = \bar{\vartheta}_g \qquad \text{for every } t. \tag{15}$$

Substitution of equation (15) into equations (9)–(11) proves the proposition. No approximation is involved.

The usual no-variation hypothesis is

$$H_0 : A_1 = \cdots = A_G = 0. \tag{16}$$

Equation (16) is a regular interior restriction only when the baseline is interior, the slope directions remain identified under the null, and the information matrix has full rank. Zero ARCH loadings, the CCC limit of DCC, a shared-index direction under zero loading, and density-family limits violate these conditions.

4 Metric-state composition

Let the joint state be the finite product

$$\mathsf{S} = \prod_{j=1}^{J} \mathsf{S}_j, \qquad \mathbf{d}(s, \widetilde{s}) = \{d_1(s_1, \widetilde{s}_1), \ldots, d_J(s_J, \widetilde{s}_J)\}'. \tag{17}$$

Each (S_j, d_j) is a complete separable metric space. Choose weights $w_j > 0$, collect them in $w = (w_1, \ldots, w_J)'$, and define

$$\|x\|_{\infty,w} = \max_{1 \le j \le J} \frac{|x_j|}{w_j}, \qquad \|L\|_{\infty,w} = \max_{1 \le j \le J} \frac{(Lw)_j}{w_j}, \qquad d_w(s, \widetilde{s}) = \|\mathbf{d}(s, \widetilde{s})\|_{\infty,w}. \tag{18}$$

For nonnegative vectors, $x \preceq y$ means componentwise inequality. We write $\log^+ x = \max\{\log x, 0\}$.

For each date t and candidate c, a component certificate supplies a nonnegative matrix $L_t(c)$ such that

$$\mathbf{d}\{\Psi_{c,t}(s), \Psi_{c,t}(\widetilde{s})\} \preceq L_t(c)\mathbf{d}(s, \widetilde{s}). \tag{19}$$

The entry $L_{jk,t}(c)$ bounds the effect of state block k on the next value of block j. Matrix multiplication therefore composes the certificates in the same order as the state updates.

Theorem 1 (Blockwise random-Lipschitz composition)**.** *Suppose $\mathcal{B}$ is compact; $\{\Psi_{c,t}\}$ is stationary and ergodic in t, jointly measurable, and almost surely continuous in (s, c); and equation (19) holds. For a fixed anchor s°, assume*

$$\mathbb{E} \log^+ \sup_{c \in \mathcal{B}} d_w\{\Psi_{c,0}(s^\circ), s^\circ\} < \infty, \tag{20}$$

$$\mathbb{E} \log^+ \sup_{c \in \mathcal{B}} \|L_0(c)\|_{\infty,w} < \infty, \tag{21}$$

$$\mathbb{E} \log \sup_{c \in \mathcal{B}} \|L_0(c)L_{-1}(c) \cdots L_{-r+1}(c)\|_{\infty,w} < 0 \quad \text{for some } r \ge 1. \tag{22}$$

Then the candidate recursion has a unique stationary causal field $c \mapsto S_t^c$ in $C(\mathcal{B}, \mathsf{S})$. Every admissible continuous seed field $c \mapsto \widetilde{S}_0^c$ forgets its initialization uniformly: for some deterministic $\rho \in (0, 1)$ and random finite $C(\omega)$,

$$\sup_{c\in\mathcal{B}} d_w(\widetilde{S}_t^c, S_t^c) \le C(\omega)\rho^t \qquad \textit{eventually almost surely}. \tag{23}$$

The proof applies random-contraction theory to the function-valued map $(\mathcal{T}_t f)(c) = \Psi_{c,t}\{f(c)\}$ [9, 34]. Applying the theorem to an augmented innovation-driven recursion proves existence of the data-generating state. Applying it to the observed-data candidate map proves filter invertibility. These are different applications; neither conclusion implies the other.

The theorem gives a precise closure rule. Block-diagonal addition preserves contraction. Feed-forward addition also preserves the exponential rate under tempered cross-gains. Cyclic coupling requires equation (22) for the full matrix. Removing a component preserves the certificate only when the remaining coordinates form an invariant subsystem. Adding or deleting an arbitrary component is not an automatic stability operation.

For a deterministic majorant $L_t(c) \le \overline{L}$, a convenient sufficient certificate is

$$\rho(\overline{L}) < 1, \qquad w = \left(I - \frac{\overline{L}}{\rho_0}\right)^{-1}\mathbf{1}, \qquad \rho(\overline{L}) < \rho_0 < 1, \tag{24}$$

because equation (24) implies $\|L_t(c)\|_{\infty,w} \le \rho_0$. This bound is auditable but may be conservative. An expected-log product condition can certify paths that no uniform one-step bound accepts.

Pointwise matrix stability is insufficient. Write $\rho(A)$ for the spectral radius of a square matrix A. For

$$A_1 = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1/2 & 0 \\ 1 & 1/2 \end{pmatrix}, \tag{25}$$

each matrix has spectral radius $1/2$, but

$$\rho(A_1A_2) = \frac{3+2\sqrt{2}}{4} > 1. \tag{26}$$

Thus even stable VAR or state-space regimes can expand under switching. The corresponding worst-case object is the joint spectral radius [46]. A stronger, distribution-free AR or VAR certificate is the common quadratic inequality

$$A_t(c)'PA_t(c) \le \rho_A^2 P, \qquad P \succ 0, \qquad 0 < \rho_A < 1, \tag{27}$$

where $\|x\|_P = (x'Px)^{1/2}$. The inequality implies $\|A_t(c)\cdots A_1(c)x\|_P \le \rho_A^t\|x\|_P$ for every admissible switching path. Equation (27) is sufficient, not necessary, and it may be a semi-infinite robust inequality when the modifier support is continuous.

5 Stationary likelihood transfer

The block theorem supplies state existence and seed forgetting. The likelihood still needs parameter geometry, moments, identification, and a score limit. A model may replace contraction with a drift, coupling, Green-function, or Markov-chain theorem that proves the same state conclusions.

5.1 Assumptions

A1. Parameter geometry. The coefficient set $\mathcal{B}$ is compact. For regular asymptotic normality, $c_\star$ lies in its interior. Each map T_g is a C^3 embedding on an open neighborhood of the essential true coordinate path. In particular,

$$\operatorname{rank} J_{T_g}\{\eta_{g,t}(c_\star)\} = r_g \qquad \text{almost surely.} \tag{28}$$

The path obeys equation (5), and the first three chart derivatives have stationary integrable envelopes. For bounded modifier support, a compact coordinate image inside U_g gives a positive uniform domain margin. For unbounded support, the model instead supplies inverse-margin or other regularity-gauge moments strong enough for the requested state and density derivatives.

A2. Predictable environment. The environment and innovation driver form a stationary ergodic process. The modifier X_t obeys equation (1). If X_t depends on past observations, the augmented data-generating state includes the variables that generate X_t.

A3. Data-generating existence. Let Z_t denote the augmented data-generating state. Suppose a stationary sequence of random maps Φ_t satisfies

$$Z_t = \Phi_t(Z_{t-1}). \tag{29}$$

For an anchor z°, assume

$$\mathbb{E}\log^+ d\{\Phi_0(z^\circ), z^\circ\} < \infty, \tag{30}$$

$$\mathbb{E}\log^+ \operatorname{Lip}(\Phi_0) < \infty, \tag{31}$$

$$\mathbb{E}\log \operatorname{Lip}(\Phi_0 \circ \cdots \circ \Phi_{-r+1}) < 0 \quad \text{for some } r \geq 1. \tag{32}$$

Equations (30)–(32) are the scalar random-contraction route [9, 13]. For a product data-generating state, the component matrices and equations (20)–(22) imply this conclusion. A model-specific drift, coupling, Green-function, or Markov-chain theorem may replace contraction.

A4. Candidate-filter invertibility. For the observed path, define the candidate update

$$\Psi_{c,t}(s) = \Psi(s, Y_{t-1:t-L}, X_{t:t-M}, \vartheta_{t:t-M}(c)). \tag{33}$$

The block composition theorem proves uniform candidate-filter forgetting when the observed-data component certificates satisfy equations (20)–(22). The following scalar majorant is the special case in which a stationary random variable Λ_t satisfies

$$\sup_{c\in\mathcal{B}} d\{\Psi_{c,t}(s), \Psi_{c,t}(s')\} \leq \Lambda_t d(s, s'), \tag{34}$$

and, for some $r \geq 1$,

$$\mathbb{E}\log^+ \sup_{c\in\mathcal{B}} d\{\Psi_{c,0}(s^\circ), s^\circ\} < \infty, \tag{35}$$

$$\mathbb{E}\log \sup_{c\in\mathcal{B}} \operatorname{Lip}(\Psi_{c,0} \circ \cdots \circ \Psi_{c,-r+1}) < 0. \tag{36}$$

Equations (35)–(36) imply uniform seed forgetting. Continuous invertibility may replace the uniform condition when it gives the likelihood, score, and Hessian approximation required below [36].

A5. Likelihood regularity. The map $(\omega, c) \mapsto q_t(\omega, c)$ is jointly measurable, and $c \mapsto q_t(c)$ is almost surely continuous on $\mathcal{B}$. The conditional density in equation (10) has parameter-independent support near $c_\star$. The negative log likelihood is twice differentiable almost surely and has the envelopes

$$\mathbb{E} \sup_{c \in \mathcal{B}} |q_t(c)| < \infty, \tag{37}$$

$$\mathbb{E} \sup_{c \in \mathcal{N}} \|\nabla_c q_t(c)\|^{2+\epsilon} < \infty, \tag{38}$$

$$\mathbb{E} \sup_{c \in \mathcal{N}} \|\nabla_c^2 q_t(c)\|^{1+\epsilon} < \infty, \tag{39}$$

and the local L^1 Hessian modulus

$$\lim_{\delta \downarrow 0} \mathbb{E} \sup_{\|c - c_\star\| \le \delta} \|\nabla_c^2 q_t(c) - \nabla_c^2 q_t(c_\star)\| = 0. \tag{40}$$

where $\epsilon > 0$ and $\mathcal{N}$ is a neighborhood of $c_\star$. A nonsmooth model may use an appropriate M-estimation theorem, but it then leaves the smooth branch stated here.

A6. Initialization is negligible. Let $\widetilde{q}_t$ use an arbitrary admissible numerical seed, and let q_t use the stationary infinite-past filter. Assume

$$\sup_{c \in \mathcal{B}} \left| \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{I}_n} \{\widetilde{q}_t(c) - q_t(c)\} \right| = o_p(1), \tag{41}$$

$$\frac{1}{\sqrt{n_{\text{eff}}}} \sum_{t \in \mathcal{I}_n} \{\nabla \widetilde{q}_t(c_\star) - \nabla q_t(c_\star)\} = o_p(1), \tag{42}$$

$$\sup_{c \in \mathcal{N}} \left\| \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{I}_n} \{\nabla^2 \widetilde{q}_t(c) - \nabla^2 q_t(c)\} \right\| = o_p(1). \tag{43}$$

Uniform filter and derivative forgetting provide a direct sufficient route to equations (41)–(43).

A7. Identification and curvature. Define the population criterion

$$Q(c) = \mathbb{E} q_0(c). \tag{44}$$

Assume

$$Q(c) > Q(c_\star) \qquad \text{for every } c \neq c_\star. \tag{45}$$

Let

$$A = \mathbb{E}\{\nabla_c^2 q_0(c_\star)\}. \tag{46}$$

Assume A is finite and nonsingular.

A8. Score limit. Let

$$g_t = \nabla_c q_t(c_\star). \tag{47}$$

Assume the central limit theorem

$$\frac{1}{\sqrt{n_{\text{eff}}}} \sum_{t \in \mathcal{I}_n} g_t \Longrightarrow N(0, B), \tag{48}$$

where B is finite and positive semidefinite. Under a correct regular conditional density, g_t is a martingale difference and $B = \mathbb{E}(g_t g_t')$. Under general misspecification, B may be the long-run covariance

$$B = \sum_{h=-\infty}^{\infty} \operatorname{Cov}(g_t, g_{t-h}). \tag{49}$$

5.2 Result

Theorem 2 (Initialized stationary likelihood estimation)**.** *Under A1–A8, the initialized criterion has the same asymptotic minimizer as the stationary infinite-past criterion. Any approximate minimizer $\widehat{c}_n$ satisfying*

$$\widetilde{Q}_n(\widehat{c}_n) \le \inf_{c \in \mathcal{B}} \widetilde{Q}_n(c) + o_p(n_{\text{eff}}) \tag{50}$$

is consistent:

$$\widehat{c}_n \xrightarrow{p} c_\star. \tag{51}$$

If $c_\star$ is interior and the numerical first-order residual satisfies

$$\|\nabla_c \widetilde{Q}_n(\widehat{c}_n)\| = o_p(\sqrt{n_{\text{eff}}}), \tag{52}$$

then

$$\sqrt{n_{\text{eff}}}(\widehat{c}_n - c_\star) \Longrightarrow N(0, A^{-1} B A^{-T}). \tag{53}$$

If the conditional density is correctly specified and regular, information equality gives

$$A = B, \qquad \operatorname{Avar}\{\sqrt{n_{\text{eff}}}(\widehat{c}_n - c_\star)\} = A^{-1}. \tag{54}$$

The theorem is a transfer result. The block theorem, a coupling argument, a Green-function result, or a Markov-chain theorem proves A3 and A4. The derivative-jet theorem below supplies a reusable route to A6. A model's density and moment theory proves A5 and A8. Identification remains a statement about the observable conditional law.

Corollary 1 (Feasible sandwich covariance)**.** *Define*

$$\widehat{A}_n = \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{I}_n} \nabla_c^2 \widetilde{q}_t(\widehat{c}_n). \tag{55}$$

If $\widehat{B}_n \xrightarrow{p} B$, then

$$\widehat{\Sigma}_n = \widehat{A}_n^{-1} \widehat{B}_n \widehat{A}_n^{-T} \xrightarrow{p} A^{-1} B A^{-T}. \tag{56}$$

Under a correctly specified regular conditional density, the score is a martingale difference and the zero-lag outer product can estimate B. Under general misspecification, a consistent long-run covariance estimator is required; a Bartlett or Newey–West estimator is one option under its bandwidth, moment, and dependence conditions [45].

Corollary 2 (Rolling finite-memory truncation)**.** *Suppose a length-m approximation has uniform per-observation score and Hessian errors bounded by an integrable envelope times ρ^m, where $0 < \rho < 1$. If*

$$\sqrt{n_{\text{eff}}}\rho^{m_n} \longrightarrow 0, \tag{57}$$

then the truncated estimator has the same first-order limit. A sufficient logarithmic choice is

$$m_n \geq \frac{(1/2 + \delta)\log n_{\text{eff}}}{|\log \rho|}, \qquad \delta > 0. \tag{58}$$

This corollary concerns rolling truncation. A filter initialized once at the start has a different, summable error calculation.

The population invertibility set may be unknown. Let $\widetilde{L}_t(c)$ be the sample block-Lipschitz matrix evaluated along the initialized observed-data filter. A feasible analogue of the block certificate starts from

$$\widehat{\lambda}_{n,r}(c) = \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{I}_n} \log \|\widetilde{L}_t(c) \cdots \widetilde{L}_{t-r+1}(c)\|_{\infty,w} \tag{59}$$

and defines the inner region

$$\widehat{\mathcal{B}}_n(\delta_n) = \{c \in \mathcal{B} : \widehat{\lambda}_{n,r}(c) \leq -\delta_n\}. \tag{60}$$

Consistency over equation (60) requires a uniform law for equation (59), $\delta_n \downarrow 0$ slowly enough to dominate its error, inclusion of $c_\star$ with probability tending to one, and identification on the limiting set. The empirical constraint certifies the candidate filter, not existence of the data-generating process. Blasques et al. prove feasible invertibility sets and confidence bounds for a broad observation-driven class [37]; extending their boundary theory to a block matrix-product statistic is separate work.

6 Exact path derivatives

6.1 Basis and chart derivatives

For block g, stack C_g by columns and define

$$K_{g,t} = \frac{\partial \eta_{g,t}}{\partial \operatorname{vec}(C_g)'} = b_g(X_t)' \otimes I_{r_g}. \tag{61}$$

Let $J_{T_g}(\eta)$ be the Jacobian of T_g. The path Jacobian is

$$D_{g,t} = \frac{\partial \vartheta_{g,t}}{\partial \operatorname{vec}(C_g)'} = J_{T_g}(\eta_{g,t}) K_{g,t}. \tag{62}$$

For component a of T_g, let $H_{T_{g,a}}(\eta)$ be its Hessian. The coefficient Hessian is

$$\nabla^2_{\operatorname{vec}(C_g)} \vartheta_{g,t,a} = K'_{g,t} H_{T_{g,a}}(\eta_{g,t}) K_{g,t}. \tag{63}$$

Equation (63) is zero only for an affine chart. Logistic, softmax, hyperspherical, correlation-matrix, root, and moment-standardization maps all contribute curvature.

Injectivity and regularity are different. The scalar map $T(\eta) = \eta^3$ is injective, but $T'(0) = 0$. It preserves set-theoretic identification while destroying ordinary first-order information at $\eta = 0$.

Equation (28) excludes this case from the regular branch.

For a finite boundary, define the path margin

$$\Delta_{g,t}(c) = \operatorname{dist}\{\vartheta_{g,t}(c), \partial\Theta_g\}. \tag{64}$$

With bounded modifier support and a compact coordinate image inside U_g, continuity gives a positive uniform lower bound for equation (64). With unbounded modifiers, a logistic, exponential, or softmax path can approach its boundary. The relevant assumption is then a model-specific inverse-margin moment, for example

$$\mathbb{E} \sup_{c \in \mathcal{N}(c_\star)} \Delta_{g,t}(c)^{-a} < \infty, \tag{65}$$

where $a > 0$ is set by the state and density derivative envelopes. A model-owned regularity gauge may replace Euclidean distance when degeneration occurs at infinity or through a smallest eigenvalue. A fitted minimum margin is useful diagnostics, but it does not establish equation (65).

6.2 State sensitivities

Write the candidate recursion as $S_t^c = \Psi_t(S_{t-1}^c, c)$. Define

$$\dot{S}_t = \frac{\partial S_t^c}{\partial c'}, \qquad \ddot{S}_t = \frac{\partial^2 S_t^c}{\partial c\, \partial c'}. \tag{66}$$

The exact first derivative recursion is

$$\dot{S}_t = \Psi_{s,t}\dot{S}_{t-1} + \Psi_{c,t}, \tag{67}$$

where $\Psi_{s,t}$ and $\Psi_{c,t}$ are partial derivatives evaluated at (S_{t-1}^c, c). For directions $u, v \in \mathbb{R}^p$, the exact second derivative recursion is

$$\begin{aligned}\ddot{S}_t[u,v] = {} & \Psi_{s,t}\ddot{S}_{t-1}[u,v] + \Psi_{ss,t}[\dot{S}_{t-1}u, \dot{S}_{t-1}v] \\ & + \Psi_{sc,t}[\dot{S}_{t-1}u, v] + \Psi_{cs,t}[u, \dot{S}_{t-1}v] + \Psi_{cc,t}[u,v].\end{aligned} \tag{68}$$

Equations (67) and (68) propagate a coefficient change through every later residual, variance, scale, and dependence state. A derivative that holds the lagged filtered state fixed differentiates another objective.

6.3 Derivative-jet stability

The metric theorem does not by itself justify equations (67) and (68). For this branch, let the state be an open subset of a separable Banach space. Suppose Ψ_t is twice Fréchet differentiable on an open tube containing the stationary path. Define

$$A_t(c) = D_s\Psi_t(S_{t-1}^c, c), \qquad B_t(c) = D_c\Psi_t(S_{t-1}^c, c). \tag{69}$$

For $j \geq 1$, write

$$A_{t:j}(c) = A_t(c)A_{t-1}(c)\cdots A_{t-j+1}(c), \qquad A_{t:0}(c) = I. \tag{70}$$

For directions $u, v \in \mathbb{R}^p$, define the second-order forcing

$$R_t(c)[u,v] = D^2_{ss}\Psi_t[\dot S_{t-1}u, \dot S_{t-1}v] + D^2_{sc}\Psi_t[\dot S_{t-1}u, v] + D^2_{cs}\Psi_t[u, \dot S_{t-1}v] + D^2_{cc}\Psi_t[u,v]. \tag{71}$$

Theorem 3 (Stationary derivative jet)**.** *Assume, uniformly over $c \in \mathcal{B}$, that the operator cocycle in equation (70) has a negative expected-log product exponent and a finite one-step $\log^+$ moment. Assume also*

$$\mathbb{E}\log^+ \sup_{c\in\mathcal{B}} \|B_0(c)\| < \infty, \qquad \mathbb{E}\log^+ \sup_{c\in\mathcal{B}} \|R_0(c)\| < \infty. \tag{72}$$

Then the unique stationary first and second coefficient derivatives are

$$\dot S_t(c) = \sum_{j=0}^{\infty} A_{t:j}(c) B_{t-j}(c), \tag{73}$$

$$\ddot S_t(c) = \sum_{j=0}^{\infty} A_{t:j}(c) R_{t-j}(c). \tag{74}$$

If $D_s\Psi_t$, $D_c\Psi_t$, and $D^2\Psi_t$ are locally Lipschitz in the state with suitable tempered envelopes, then initialized derivatives forget their seeds at a possibly slower exponential rate: for some $\bar\rho \in (0,1)$,

$$\sup_{c\in\mathcal{B}} \|\widetilde{\dot S}_t(c) - \dot S_t(c)\| \le C_1(\omega)\bar\rho^t, \tag{75}$$

$$\sup_{c\in\mathcal{B}} \|\widetilde{\ddot S}_t(c) - \ddot S_t(c)\| \le C_2(\omega)\bar\rho^t. \tag{76}$$

The theorem is sequential. First prove the state, then the first tangent, then the second tangent. The forcing in equation (71) is quadratic in $\dot S$, so the unrestricted joint update $(S, \dot S, \ddot S)$ is not globally Lipschitz. Log moments give existence; score and Hessian inference may require stronger L^q Green-product and forcing moments. Under those moments and regularity of q_t in the state, equations (75) and (76) imply the score and Hessian initialization transfers in A6 [34, 38].

6.4 Likelihood pullback

Let $V_t(c) \in \mathbb{R}^d$ collect the local density inputs, including the current residual, scale, shape, and dependence coordinates. Write

$$q_t(c) = \bar q_t\{V_t(c)\}. \tag{77}$$

Let $V_{t,c}$ be the Jacobian of V_t, and let $V_{t,j,cc}$ be the Hessian of its j-th component. The exact score and Hessian are

$$\nabla_c q_t = V'_{t,c}\nabla_V \bar q_t, \tag{78}$$

$$\nabla^2_c q_t = V'_{t,c}\nabla^2_V \bar q_t V_{t,c} + \sum_{j=1}^{d} \frac{\partial \bar q_t}{\partial V_{t,j}} V_{t,j,cc}. \tag{79}$$

The second term in equation (79) contains chart curvature and recursive-state curvature. It is part of the exact Hessian.

An equivalent full-path statement is useful for validation. Let $\Theta(c)$ stack the parameter path

and let J_Θ be its Jacobian. If g_Θ and H_Θ are the value, gradient, and Hessian of the complete path likelihood, then

$$\nabla_c Q_n = J'_\Theta g_\Theta, \tag{80}$$

$$\nabla_c^2 Q_n = J'_\Theta H_\Theta J_\Theta + \sum_j g_{\Theta,j} \nabla_c^2 \Theta_j. \tag{81}$$

Equation (81) shows why a fixed-parameter jet followed by a first-order chain rule is insufficient. The algebra extends to a smooth additive conditional contrast

$$q_t(c) = \rho_t\{Y_t, V_t(c), c\}. \tag{82}$$

If $J_t = D_c V_t$ and $H_{t,j} = D_c^2 V_{t,j}$, then

$$\nabla_c q_t = J'_t \rho_{V,t} + \rho_{c,t}, \tag{83}$$

$$\nabla_c^2 q_t = J'_t \rho_{VV,t} J_t + \sum_j \rho_{V_j,t} H_{t,j} + J'_t \rho_{Vc,t} + \rho_{cV,t} J_t + \rho_{cc,t}. \tag{84}$$

The stationary M-estimation transfer applies if the contrast separately satisfies its ULLN, identification, smoothness, and score-limit conditions. The statistical conclusions do not transfer unchanged. Strict propriety identifies a predictive law rather than a coefficient vector; quantile and expectile losses leave the ordinary C^2 branch; generic GMM is a quadratic function of sample-average moments rather than an additive criterion; and composite likelihood generally uses Godambe rather than Fisher information. Observation-driven moment and proper-score filters are established but candidate-dependent state models [40, 44]. For these reasons, the main theorem remains a conditional-likelihood theorem.

7 Identification after recursive composition

7.1 The basis layer

Define the basis Gram matrix for block g by

$$G_g = \mathbb{E}\{b_g(X_t) b_g(X_t)'\}. \tag{85}$$

If G_g is positive definite, T_g is injective, and the baseline normalization removes every exact symmetry, then

$$C_g b_g(X_t) = \widetilde{C}_g b_g(X_t) \quad \text{almost surely} \quad \Longrightarrow \quad C_g = \widetilde{C}_g. \tag{86}$$

Equation (86) identifies the coefficient function on the support of X_t. Values outside that support are extrapolations imposed by the basis and chart.

7.2 The observable-law layer

Let $\mathcal{P}$ map an admissible parameter path into the observable law generated by equations (9) and (10). Let Φ map c into its parameter path. Global identification requires

$$\mathcal{P}\{\Phi(c)\} = \mathcal{P}\{\Phi(\widetilde{c})\} \quad \Longrightarrow \quad c = \widetilde{c}. \tag{87}$$

Local identification requires

$$\ker D\mathcal{P}_{\Phi(c_\star)} \cap \operatorname{im} D\Phi_{c_\star} = \{0\}. \tag{88}$$

In a regular likelihood model, equation (88) is equivalent to nonsingularity of the pulled-back Fisher information under the conditions studied by Rothenberg [31]. If $\mathcal{I}_\Theta$ denotes path information, then

$$\mathcal{I}_c = D\Phi_{c_\star}^* \mathcal{I}_\Theta D\Phi_{c_\star}. \tag{89}$$

Full rank of equation (85) does not imply full rank of equation (89). A recursive state can make a basis direction observationally silent. Dynamic cancellation, a zero ARCH loading, a redundant softmax coordinate, or a vanishing shared-index loading can all cause this failure.

Let $\lambda_t(c)$ denote the complete C_t-measurable predictive-density parameter before Y_t is observed. It may contain location, scale, shape, and correlation coordinates, but not the realized current residual. Define

$$D_t = \frac{\partial \lambda_t(c_\star)}{\partial c'}, \qquad u_t = \frac{\partial \log p(Y_t \mid C_t; \lambda_t)}{\partial \lambda_t}, \qquad \mathcal{I}_{\lambda,t} = \mathbb{E}(u_t u_t' \mid C_t). \tag{90}$$

The Jacobian D_t includes every direct chart derivative and every derivative propagated through the stationary tangent recursion.

Proposition 2 (Recursive information rank)**.** *Under correct regular conditional specification and dominated differentiation,*

$$\mathcal{I}_c = \mathbb{E}(D_t' \mathcal{I}_{\lambda,t} D_t), \qquad h' \mathcal{I}_c h = \mathbb{E}\left\| \mathcal{I}_{\lambda,t}^{1/2} D_t h \right\|^2. \tag{91}$$

Consequently, $\mathcal{I}_c > 0$ if and only if

$$\mathcal{I}_{\lambda,t}^{1/2} D_t h = 0 \quad \textit{almost surely} \quad \Longrightarrow \quad h = 0. \tag{92}$$

At regular points, equation (92) is a local identification certificate. It does not prove global identification. Under misspecification, the outer product is not an identification certificate because information equality fails; curvature A and score variation B remain distinct.

An empirical diagnostic is

$$\widehat{\mathcal{I}}_{c,n} = \frac{1}{n_{\text{eff}}} \sum_{t \in \mathcal{I}_n} \widehat{D}_t' \widehat{\mathcal{I}}_{\lambda,t} \widehat{D}_t. \tag{93}$$

Software can report singular values, condition numbers, and the weak coefficient directions of equation (93). These are diagnostics, not population proofs. The singular values of the stacked weighted sensitivities are numerically preferable to forming the normal matrix.

For a correctly specified conditional density, conditional Kullback–Leibler divergence gives a direct identification route:

$$\mathbb{E}\{q_t(c) - q_t(c_0) \mid C_t\} = \mathrm{KL}\big(p_{c_0,t} \,\|\, p_{c,t}\big) \geq 0. \tag{94}$$

Equality in equation (94), together with equation (87), identifies c_0.

7.3 Regular test of no coefficient variation

Let R select the nonconstant basis coefficients, let $r = \operatorname{rank}(R)$, and let

$$\Sigma = A^{-1}BA^{-T} \tag{95}$$

denote the covariance of the root-n_{eff} limit. Suppose zero slopes are interior, all nuisance parameters and slope directions remain identified under $H_0 : Rc_\star = 0$, and $R\Sigma R' \succ 0$. Then

$$W_n = n_{\text{eff}}(R\widehat{c}_n)'(R\widehat{\Sigma}_n R')^{-1}(R\widehat{c}_n) \Longrightarrow \chi^2_r. \tag{96}$$

Under regular local alternatives $Rc_n = h/\sqrt{n_{\text{eff}}}$,

$$W_n \Longrightarrow \chi^2_r\left[h'(R\Sigma R')^{-1}h\right]. \tag{97}$$

Equation (96) defines $\widehat{\Sigma}_n$ as an estimator of the root-n_{eff} covariance. If software reports $\widehat{\Sigma}_n/n_{\text{eff}}$, the leading sample-size factor is omitted. A score test can fit only the fixed model, but it must still propagate every slope tangent through the recursion. Boundary points, singular charts, and nuisance parameters that disappear under H_0 require model-specific cone or Gaussian-process limits [41, 32, 43].

8 Scope boundaries and next theorems

8.1 Stationary predictable variation

The theorem in Section 5 treats a stationary ergodic environment. Examples include an external stationary regressor, a stationary regime indicator, and a predetermined state included in the augmented recursion. A correctly specified finite-dimensional path has the root-n limit in equation (53).

8.2 Deterministic finite-dimensional paths are triangular arrays

Let $X_{n,t} = x(t/n)$ for a bounded deterministic function $x : [0,1] \to \mathbb{R}^k$. Define

$$\vartheta_{n,t}(c) = T\{Cb(x(t/n))\}. \tag{98}$$

This specification forms a triangular array. Stationary ergodic laws no longer prove consistency. A finite-dimensional theorem instead requires a uniform array law of large numbers,

$$\sup_{c\in\mathcal{B}}\left|\frac{1}{n}\sum_{t=1}^{n} q_{n,t}(c) - Q(c)\right| \xrightarrow{p} 0, \tag{99}$$

a martingale-array or dependent-array central limit theorem,

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n} g_{n,t}(c_\star) \Longrightarrow N(0,B), \tag{100}$$

and average Hessian convergence,

$$\frac{1}{n}\sum_{t=1}^{n} \nabla_c^2 q_{n,t}(c_\star) \xrightarrow{p} A. \tag{101}$$

Under these separately verified conditions, the root-n sandwich limit remains valid. Calendar

time should enter as t/n, or through another controlled basis. Raw unscaled time can make predictors and information diverge. The stationary theorem does not prove equations (99)–(101).

8.3 Nonparametric coefficient curves

Suppose the coefficient curve $\vartheta(u)$ is estimated locally at $u \in (0,1)$ with bandwidth h. The effective local sample size is

$$n_{\text{local}} \asymp nh. \tag{102}$$

For a regular fixed-dimensional local-linear problem, the benchmark pointwise expansion is

$$\widehat{\vartheta}(u) - \vartheta(u) = O_p\{(nh)^{-1/2}\} + O(h^2), \tag{103}$$

Equation (103) is a benchmark, not a consequence of the stationary theorem. It requires a twice differentiable curve, kernel and bandwidth conditions, local identification, curvature, dependence control, and a stationary approximation to the time-varying recursion. Uniform bands need stronger conditions and logarithmic normalization [12, 4, 21, 42]. This is a separate estimator from the finite-dimensional global basis in equation (98).

A growing-sieve branch is also separate. If the coefficient dimension p_n increases, stability, derivative envelopes, information conditioning, empirical-process complexity, approximation error, and penalty bias must hold uniformly in p_n. Generic sieve time-series inference already treats regular and irregular functionals [39]. The missing contribution would be uniform recursive-state and chart feasibility, not the generic rate $\sqrt{p_n/n}$. Neither the locally stationary nor the sieve branch is proved here.

8.4 Latent stochastic coefficients

A latent coefficient state obeys another model:

$$A_t \mid A_{t-1}, X_t \sim K_c(\cdot \mid A_{t-1}, X_t), \tag{104}$$

$$Y_t \mid A_t, \mathcal{F}_{t-1} \sim p_c(\cdot \mid A_t). \tag{105}$$

The observed predictive density integrates over the filtering distribution $\pi^c_{t|t-1}$:

$$p_c(Y_t \mid \mathcal{F}_{t-1}, X_{\le t}) = \int p_c(Y_t \mid a)\, \pi^c_{t|t-1}(da). \tag{106}$$

In general,

$$\int p_c(y \mid a)\, \pi(da) \neq p_c\{y \mid \mathbb{E}_\pi(A)\}. \tag{107}$$

Thus a latent time-varying parameter model uses a probability-measure-valued filter and an integrated likelihood. It needs partially observed Markov or state-space theory. It should not share an inference claim with the predictable path in equation (4).

9 Two worked certificates

9.1 Gaussian ARX and HARX

Let $b_t = b(X_t) \in \mathbb{R}^k$ be a bounded predictable basis vector, let $W_t \in \mathbb{R}^d$ be a predictable regressor, and define the lag vector $s_{t-1} = (Y_{t-1}, \dots, Y_{t-p})'$. For $c = \mathrm{vec}(C)$, partition

$$\begin{pmatrix} \phi_t(c) \\ \delta_t(c) \end{pmatrix} = Cb_t, \qquad \phi_t(c) \in \mathbb{R}^p, \qquad \delta_t(c) \in \mathbb{R}^d. \tag{108}$$

The model is

$$Y_t = \phi_t(c_\star)' s_{t-1} + \delta_t(c_\star)' W_t + \sigma Z_t, \qquad Z_t \overset{\text{iid}}{\sim} N(0,1), \qquad \sigma > 0. \tag{109}$$

Let $A(\phi)$ be the $p \times p$ companion matrix whose first row is ϕ'. Suppose $\mathbb{E}\|W_t\|^{4+\epsilon} < \infty$ and, uniformly over the compact coefficient domain,

$$A\{\phi_t(c)\}' P A\{\phi_t(c)\} \preceq \rho^2 P, \qquad P \succ 0, \qquad 0 < \rho < 1. \tag{110}$$

With $e_1 = (1, 0, \dots, 0)'$, the data-generating state satisfies

$$s_t = A\{\phi_t(c_\star)\} s_{t-1} + e_1 \{\delta_t(c_\star)' W_t + \sigma Z_t\}. \tag{111}$$

Equation (110) gives $\|A_t \cdots A_{t-j+1}\|_P \le \rho^j$, hence a unique stationary causal state with the required moment. For a parameter direction v, the law-state tangent obeys

$$\dot{s}_t[v] = A_t \dot{s}_{t-1}[v] + e_1 \{\dot{\phi}_t[v]' s_{t-1} + \dot{\delta}_t[v]' W_t\}, \tag{112}$$

with the same stable homogeneous matrix. The second tangent has the same linear part and the forcing from equation (71). Because equation (109) uses observed lags, the conditional likelihood itself has no candidate filter.

Define

$$u_t = \begin{pmatrix} s_{t-1} \\ W_t \end{pmatrix}, \qquad r_t = b_t \otimes u_t, \qquad m_t(c) = u_t' C b_t. \tag{113}$$

The exact negative log likelihood and its first two derivatives are

$$q_t(c) = \frac{1}{2} \left\{ \log(2\pi\sigma^2) + \frac{[Y_t - m_t(c)]^2}{\sigma^2} \right\}, \tag{114}$$

$$\nabla_c q_t(c) = -\frac{Y_t - m_t(c)}{\sigma^2} r_t, \tag{115}$$

$$\nabla_c^2 q_t(c) = \frac{1}{\sigma^2} r_t r_t'. \tag{116}$$

The rank certificate is

$$\mathcal{I}_{\mathrm{ARX}} = \frac{1}{\sigma^2} \mathbb{E}(r_t r_t') \succ 0. \tag{117}$$

It gives local information rank and, because equal Gaussian conditional laws have equal means, global coefficient identification. The estimation theorem yields

$$\sqrt{n-p}(\widehat{c}_n - c_\star) \Longrightarrow N(0, \mathcal{I}_{\mathrm{ARX}}^{-1}). \tag{118}$$

HARX is a linear restriction of ARX. For horizons $h_1, \dots, h_K$, let $H = \max_k h_k$ and define

$$[M_H]_{j,k} = \frac{1(j \le h_k)}{h_k}, \qquad \phi_t = M_H \beta_t. \tag{119}$$

Equations (110)–(118) apply to $A(M_H \beta_t)$. HARX changes the linear restriction and information design, not the probability theorem.

9.2 Covariate-indexed GARCH-X(1,1)

Let $b_t = (1, \widetilde{b}(X_t)')'$, where X_t is observed before Y_t. Assume X_t is iid with bounded support and independent of iid $Z_t \sim N(0,1)$, and assume

$$G_b = \mathbb{E}(b_t b_t') \succ 0. \tag{120}$$

Define

$$Y_t = \sqrt{h_t} Z_t, \tag{121}$$
$$h_t = \omega_t + \alpha Y_{t-1}^2 + \beta h_{t-1}, \tag{122}$$
$$\omega_t = \exp(c_\omega' b_t), \tag{123}$$
$$(\alpha, \beta, 1-\alpha-\beta) = \frac{(e^{u_\alpha}, e^{u_\beta}, 1)}{1 + e^{u_\alpha} + e^{u_\beta}}. \tag{124}$$

The coefficient vector is $c = (c_\omega', u_\alpha, u_\beta)'$. Compactness of its regular domain and bounded b_t give

$$0 < \underline{\omega} \le \omega_t(c) \le \overline{\omega} < \infty, \qquad 0 < \beta(c) \le \overline{\beta} < 1. \tag{125}$$

At the truth, the data-generating recursion is

$$h_t = \omega_t + A_t h_{t-1}, \qquad A_t = \alpha Z_{t-1}^2 + \beta. \tag{126}$$

Strict stationarity follows from $\mathbb{E} \log A_t < 0$. The simplex chart gives the sufficient bound $\mathbb{E} \log A_t \le \log(\alpha + \beta) < 0$. To obtain the likelihood derivative moments, impose for some $\epsilon > 0$

$$s = 2(1+\epsilon), \qquad \mathbb{E}(A_t^s) < 1, \qquad \mathbb{E}|Z_t|^{2s} < \infty. \tag{127}$$

The observed candidate filter has the different contraction

$$|\widetilde{h}_t(c) - h_t(c)| \le \overline{\beta}^t |\widetilde{h}_0(c) - h_0(c)|. \tag{128}$$

For a direction v, let $d_t[v] = \partial h_t(c)[v]$. The exact first tangent is

$$d_t[v] = \dot{\omega}_t[v] + \dot{\alpha}[v] Y_{t-1}^2 + \dot{\beta}[v] h_{t-1} + \beta d_{t-1}[v]. \tag{129}$$

For directions u, v, the second tangent is

$$\begin{aligned} D_t[u,v] = {} & \ddot{\omega}_t[u,v] + \ddot{\alpha}[u,v] Y_{t-1}^2 + \ddot{\beta}[u,v] h_{t-1} \\ & + \dot{\beta}[u] d_{t-1}[v] + \dot{\beta}[v] d_{t-1}[u] + \beta D_{t-1}[u,v]. \end{aligned} \tag{130}$$

Equations (127) and (128) give stationary derivative jets and seed forgetting. The Gaussian negative log likelihood is

$$q_t(c) = \frac{1}{2}\left\{\log(2\pi) + \log h_t(c) + \frac{Y_t^2}{h_t(c)}\right\}. \tag{131}$$

Let $d_t = \nabla_c h_t(c_\star)$ and $D_t = \nabla_c^2 h_t(c_\star)$. Then

$$g_t = \frac{1}{2}(1 - Z_t^2)\frac{d_t}{h_t}, \tag{132}$$

$$H_t = \frac{1}{2}(1 - Z_t^2)\frac{D_t}{h_t} + \frac{1}{2}(2Z_t^2 - 1)\frac{d_t d_t'}{h_t^2}. \tag{133}$$

The recursive information matrix is

$$\mathcal{I}_{\mathrm{GX}} = \frac{1}{2}\mathbb{E}\left(\frac{d_t d_t'}{h_t^2}\right). \tag{134}$$

It is positive definite when Z_t^2 and h_t are nondegenerate, equation (120) holds, and the reference-softmax Jacobian has full rank. Indeed, $d_t' v = 0$ and equation (129) imply

$$\omega_t b_t' v_\omega + \dot{\alpha}[v] Y_{t-1}^2 + \dot{\beta}[v] h_{t-1} = 0 \qquad \text{almost surely.} \tag{135}$$

Conditional variation in Z_{t-1}^2 identifies α; independent variation in X_t and h_{t-1} identifies β and c_ω. The same argument gives global identification from equality of Gaussian variance paths. Hence

$$\sqrt{n-1}(\widehat{c}_n - c_\star) \Longrightarrow N(0, \mathcal{I}_{\mathrm{GX}}^{-1}). \tag{136}$$

With non-Gaussian innovations whose conditional mean and variance are correct, equation (136) uses the Gaussian-QMLE sandwich instead. If d_ω collects the nonconstant entries of c_ω, the null $H_0 : d_\omega = 0$ is regular when equation (134) has full rank. This differs from testing a constrained nonnegative additive GARCH-X loading at zero.

10 How the remaining model families enter

Table 1 records the state coordinate and the proof obligation for each family. The common theorem applies after the family verifies the listed obligation.

Table 1: Model-family embeddings and their distinct certificates

Family	State used by the common recursion	Certificate that remains family-specific
ARMA	Observation and innovation companion states	Negative Lyapunov exponent or uniform product stability for AR causality; separate MA-filter product contraction; coprimeness and path identification
ARX, HARX	AR companion state with exogenous regressors; HARX is a restricted high-order ARX	Timing and exogeneity of regressors; stability of the effective lag coefficients; full information rank

Family	State used by the common recursion	Certificate that remains family-specific
GARCH	Positive variance or variance-companion state	Positive recursion, innovation-driven SRE contraction, observed-filter contraction, and requested moment contraction
GJR-GARCH	Positive variance state with a sign-dependent shock term	Joint positivity of α_t and $\alpha_t + \gamma_t$; density-dependent negative-half moment; threshold differentiability
EGARCH	Log-variance state	Strict stationarity, continuous observed-filter invertibility, and exponential moments are separate conditions
APARCH	Powered-scale state σ_t^δ	Fixed power for the standard recursion; asymmetric power moments; differentiability at zero; a newly defined state law if the power varies
Normal, Student-t, skew-t, GED	Standardized conditional-density policy	Exact normalization; finite variance; shape-domain map; score moments; request-specific differentiability
GH, Hyperbolic, NIG, VG	Native standardized density stratum	Bessel or mixture normalization; identified chart; boundary-stratum treatment; moment and Hessian validity
CCC-X	No dependence state; correlation is a predictable chart output	Positive-definite correlation chart and eigenvalue or integrability conditions
DCC and cDCC	Positive-definite matrix state normalized to a correlation matrix	Cone preservation, data-generating stability, observed-filter stability, targeting, and first-stage uncertainty
Conditional copula	Marginal states plus an optional copula-parameter state	Valid marginal CDFs, valid copula parameter domain, joint identification, and generated-margin inference
Composite model	Product of mean, volatility, density, and dependence states	Joint rather than staged derivatives; every cross block; a common scored sample and timing plan
Automatic selection	Discrete candidate index, not a smooth state	Post-selection theory; it lies outside the fixed-dimensional theorem

The table supports one library abstraction. It does not claim that every family has the same domain, state coordinate, or contraction coefficient.

11 Implications for a general library

The mathematics suggests one reusable parameter-path layer. It should own the basis, normalization, coefficient matrices, block charts, and future modifier scenarios. Each model component should own its state coordinate, path-admissible set, filter, density, derivative validity, moments, and identification statement.

The first statistically complete implementation should use a finite-dimensional basis and observed predictable modifiers. It should distinguish the stationary and deterministic-array regimes. It should use exact path derivatives from equations (67)–(79). It should report the effective chart, covariate support, path margins, stability certificate, and information rank.

The following claims are valid:

1. Every interior fixed model is a constant-path restriction within its own family.
2. Affine and finite-spline paths use the same likelihood and derivative machinery.
3. A common transfer theorem gives consistency and sandwich inference after the model certificates hold.
4. Mean, volatility, density shape, and dependence can enter one joint augmented state.

The following claims are invalid:

1. Pointwise admissibility implies path stability.
2. Strict stationarity implies observed-filter invertibility.
3. A fixed-parameter Hessian lifts through a first-order chain rule.
4. Every supported density is twice differentiable at every data-parameter point.
5. Current two-stage DCC is an ordinary member of a joint MLE theorem.
6. A varying APARCH power is defined by replacing δ with δ_t.
7. Automatic model selection has the same asymptotic law as a fixed candidate.

The right general claim is exact and limited: a predictable parameter path, a recursive state, and a conditional law share one estimation theory when each component supplies its mathematical certificate.

12 Disposition of the proposed extensions

Table 2 separates repairs to this theorem from separate research programs. "Defer" means that the idea is valid but not a corollary of the results proved here.

Table 2: Mathematical disposition of the reviewer proposals

Proposal	Decision	Reason
Blockwise composition	Accept with limits	The weighted matrix-product theorem proves joint-state and filter closure. Arbitrary cyclic addition still requires a new full certificate.
Derivative-jet theorem	Accept	Metric state stability does not imply derivative stability. The Banach theorem and forcing moments are separate.
Stationary likelihood repairs	Accept	The initialized criterion, ULLN conditions, integrated Hessian, numerical score residual, and feasible covariance are necessary.
Empirical invertibility set	Cite and defer	Feasible regions require a uniform law and boundary inference for the empirical contraction statistic; this is established prior work for observation-driven filters [37].
Immersive charts and margins	Accept with limits	Full differential rank is needed for regular inference. Uniform margins are appropriate for bounded support; unbounded modifiers need model-specific inverse-margin moments.
Recursive information rank	Accept locally	The weighted sensitivity Gram matrix certifies regular local information rank, not global identification.

Proposal	Decision	Reason
No-variation testing	Accept regular branch	Wald and score tests are standard under interior identified slopes. Boundaries and unidentified nuisance parameters need model-specific limits [41, 43].
Score-driven modifiers	Clarify scope	Candidate-dependent GAS filters are state components. Generic score-driven MLE theory already exists [38].
Arbitrary conditional contrasts	Defer	Smooth additive contrasts share the chain rule, but proper scores change the target, quantile losses are nonsmooth, GMM is generally nonadditive, and composite likelihood lacks ordinary information equality [40, 44].
Generated-input theorem	Keep as lemma	The existing two-step influence expansion is standard under a root-n, asymptotically linear first stage. Slower or nonparametric first stages need another theorem.
Local-stationary or sieve completion	Defer both	Each branch needs its own approximation, bias, dependence, dimension, and uniform-stability theory [21, 42, 39].
Title and literature position	Accept	"Composition" names the new calculation. "Grand" would overstate the family-specific parts.
Complete model examples	Accept two	ARX/HARX tests switching stability; GARCH-X tests distinct data-generating and observed-filter contractions.

A Proof outlines

A.1 Existence and initialization forgetting

Under equations (30)–(32), the backward compositions

$$Z_t^{(k)} = \Phi_t \circ \Phi_{t-1} \circ \cdots \circ \Phi_{t-k+1}(z^\circ) \tag{137}$$

form an almost surely Cauchy sequence. Its limit $Z_t^\star$ is measurable with respect to the infinite past, stationary, and satisfies equation (29). Average contraction makes this solution unique and makes every forward trajectory forget its starting point [9, 13].

For the product filter, equations (19) and (18) give a scalar Lipschitz modulus bounded by the matrix-product norm in equation (22). The same backward-composition argument on $C(\mathcal{B}, \mathsf{S})$ gives

$$\sup_{c \in \mathcal{B}} d(\widetilde{S}_t^c, S_t^c) \le C(\omega)\rho^t \tag{138}$$

eventually, for random finite $C(\omega)$ and $0 < \rho < 1$. Likelihood continuity gives the value transfer in equation (41). It does not give the score or Hessian transfers. Those follow only after equations (75) and (76), together with the likelihood derivative envelopes, establish derivative-filter forgetting.

A.2 Uniform convergence and consistency

Stationarity, ergodicity, continuity, and equation (37) give the uniform law

$$\sup_{c\in\mathcal{B}}\left|\frac{1}{n_{\text{eff}}}Q_n(c)-Q(c)\right|\xrightarrow{p}0. \tag{139}$$

Equation (41), together with (45), (50), and (139), transfers the initialized criterion to the stationary criterion and implies equation (51) by the standard argmin theorem.

A.3 Asymptotic linearization

For an interior approximate numerical minimizer, define $r_n=\nabla_c\widetilde{Q}_n(\widehat{c}_n)$. Equation (52) gives

$$\frac{r_n}{\sqrt{n_{\text{eff}}}}=o_p(1). \tag{140}$$

The fundamental theorem of calculus gives the valid multivariate expansion

$$\begin{aligned}\frac{r_n}{\sqrt{n_{\text{eff}}}}&=\frac{1}{\sqrt{n_{\text{eff}}}}\nabla_c\widetilde{Q}_n(c_\star)\\&\quad+\left[\int_0^1\frac{1}{n_{\text{eff}}}\nabla_c^2\widetilde{Q}_n\{c_\star+s(\widehat{c}_n-c_\star)\}\,ds\right]\sqrt{n_{\text{eff}}}(\widehat{c}_n-c_\star),\end{aligned} \tag{141}$$

Assumptions A5–A7 and equation (40) make the integrated matrix converge to A. Equations (42), (48), and (52), followed by Slutsky's theorem, give equation (53). A single intermediate point common to every component of a vector score is not generally available.

A.4 Information equality

Under a correctly specified regular density, a common support and an integrable dominating envelope for the first two density derivatives justify differentiation under the conditional integral and give

$$\mathbb{E}\{g_t\mid C_t\}=0. \tag{142}$$

A second differentiation of the normalization identity yields

$$\mathbb{E}\{\nabla_c^2 q_t(c_0)\mid C_t\}=\mathbb{E}\{g_t g_t'\mid C_t\}. \tag{143}$$

Taking expectations proves equation (54). Under misspecification, White's sandwich theory applies after the dependent score limit is established [35]. Gaussian QMLE has a special conditional-mean and conditional-variance robustness result under the assumptions of Bollerslev and Wooldridge [7]. Shape-path estimates do not inherit that robustness automatically.

B Family certificates in equations

B.1 ARMA, ARX, and HARX

Define a time-varying ARMA(p, q) process by

$$Y_t = c_t + \sum_{i=1}^{p} \phi_{i,t} Y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_{j,t} \varepsilon_{t-j}. \tag{144}$$

Let A_t be the $p \times p$ AR companion matrix with first row $(\phi_{1,t}, \ldots, \phi_{p,t})$. Let D_t be the $q \times q$ innovation-filter companion matrix with first row $(-\theta_{1,t}, \ldots, -\theta_{q,t})$. AR causality concerns products of A_t:

$$\lambda_A = \lim_{n\to\infty} \frac{1}{n} \log \|A_t A_{t-1} \cdots A_{t-n+1}\| < 0. \tag{145}$$

MA invertibility concerns different products:

$$\lambda_D = \lim_{n\to\infty} \frac{1}{n} \log \|D_t D_{t-1} \cdots D_{t-n+1}\| < 0. \tag{146}$$

Pointwise roots do not control the products in equations (145) and (146). Noncommuting matrices can each be Schur stable while their products expand. The Green-function results of Karanasos, Paraskevopoulos, Magdalinos, and Canepa provide another exact ARMA path representation and family-specific summability conditions [25].

An ARX model adds a predictable regressor term $\delta_t' W_t$ to equation (144), where W_t is observed before Y_t. HARX is a restricted ARX with averages

$$\overline{Y}_{t,h_k} = \frac{1}{h_k} \sum_{j=1}^{h_k} Y_{t-j}, \qquad Y_t = c_t + \sum_{k=1}^{K} \beta_{k,t} \overline{Y}_{t,h_k} + \delta_t' W_t + \varepsilon_t. \tag{147}$$

Equation (147) is an ARX of order $H = \max_k h_k$ with effective coefficient

$$\alpha_{j,t} = \sum_{k:j\le h_k} \frac{\beta_{k,t}}{h_k}. \tag{148}$$

The companion matrices built from equation (148) own the stability certificate.

B.2 GARCH, GJR-GARCH, EGARCH, and APARCH

Let $z_t = \varepsilon_t/\sigma_t$ be the standardized innovation. First-order GARCH has the random affine variance recursion

$$h_{t+1} = \omega_{t+1} + (\alpha_{t+1} z_t^2 + \beta_{t+1}) h_t, \qquad h_t = \sigma_t^2. \tag{149}$$

A standard strict-stationarity certificate is

$$\mathbb{E} \log(\alpha_t z_{t-1}^2 + \beta_t) < 0, \qquad \mathbb{E} \log^+ \omega_t < \infty. \tag{150}$$

The observed filter replaces $z_t^2 h_t$ with ε_t^2:

$$\widehat{h}_{t+1} = \omega_{t+1} + \alpha_{t+1} \varepsilon_t^2 + \beta_{t+1} \widehat{h}_t. \tag{151}$$

The filter contraction in equation (151) depends on products of β_t, while the latent-state contraction in equation (150) depends on $\alpha_t z_{t-1}^2 + \beta_t$. These are distinct claims [8, 34].

GJR-GARCH replaces the random coefficient in equation (149) with

$$A_t^{\mathrm{GJR}} = \beta_t + \{\alpha_t + \gamma_t 1(z_{t-1} < 0)\} z_{t-1}^2. \tag{152}$$

Its variance-domain map must impose $\alpha_t \geq 0$ and $\alpha_t + \gamma_t \geq 0$. Its second-moment restriction depends on

$$\kappa_{-,t} = \mathbb{E}\{z_t^2 1(z_t < 0) \mid C_t\}. \tag{153}$$

If skewness varies, $\kappa_{-,t}$ varies with the density path. Replacing it by $1/2$ is valid only under the corresponding symmetry and standardization assumptions.

EGARCH uses the log-variance state $s_t = \log h_t$:

$$s_{t+1} = \omega_{t+1} + \beta_{t+1} s_t + \alpha_{t+1}\{|z_t| - m_{1,t}\} + \gamma_{t+1} z_t, \tag{154}$$

where

$$m_{1,t} = \mathbb{E}(|z_t| \mid C_t). \tag{155}$$

The observed EGARCH filter has derivative

$$\frac{\partial \widehat{s}_{t+1}}{\partial \widehat{s}_t} = \beta_{t+1} - \frac{1}{2}\{\alpha_{t+1}|\varepsilon_t| + \gamma_{t+1}\varepsilon_t\} \exp(-\widehat{s}_t/2). \tag{156}$$

Strict stationarity of equation (154) does not imply contraction of equation (156). EGARCH therefore requires a continuous-invertibility certificate [33, 36]. If the density shape varies, equation (155) must use the shape attached to the date when z_t was generated.

For fixed power $\delta > 0$, APARCH uses the state $u_t = \sigma_t^\delta$:

$$u_{t+1} = \omega_{t+1} + \alpha_{t+1}(|\varepsilon_t| - \gamma_{t+1}\varepsilon_t)^\delta + \beta_{t+1} u_t. \tag{157}$$

This is a positive affine observed filter. A varying power changes the lag coordinate because

$$\sigma_{t-j}^{\delta_t} = u_{t-j}^{\delta_t/\delta_{t-j}}. \tag{158}$$

Recursing directly on u_{t-j} instead of equation (158) defines another model. Varying-power APARCH therefore needs a new law of motion rather than a parameter substitution [14].

B.3 Standardized conditional densities

Let $q_k(x;\psi)$ be a native density from family k, with finite mean $m_k(\psi)$ and finite positive variance $v_k(\psi)$. Its unit-variance standardized density is

$$f_k(z;\psi) = \sqrt{v_k(\psi)}\, q_k\Big(m_k(\psi) + \sqrt{v_k(\psi)} z; \psi\Big). \tag{159}$$

The conditional observation equation is

$$Y_t = \mu_t + \sigma_t Z_t, \qquad Z_t \mid C_t \sim f_k(\cdot;\psi_t), \qquad \psi_t = T_k(C_k b_k(X_t)). \tag{160}$$

Exact standardization requires

$$\int f_k(z;\psi)\, dz = 1, \qquad \int z f_k(z;\psi)\, dz = 0, \qquad \int z^2 f_k(z;\psi)\, dz = 1. \tag{161}$$

The log likelihood contribution is

$$\ell_t = -\log \sigma_t + \log f_k\left(\frac{Y_t - \mu_t}{\sigma_t}; \psi_t\right). \tag{162}$$

When ψ_t varies, derivatives of m_k, v_k, every normalizing constant, and every parameter-dependent breakpoint belong in equation (162). Hansen's conditional-density construction is the direct precedent [22].

The Student-t unit-variance density illustrates the moment gate:

$$f_t(z;\nu) = \frac{\Gamma\{(\nu+1)/2\}}{\Gamma(\nu/2)\sqrt{\pi(\nu-2)}}\left(1 + \frac{z^2}{\nu-2}\right)^{-(\nu+1)/2}, \qquad \nu > 2. \tag{163}$$

Its absolute moment of order r exists only for $r < \nu$:

$$\mathbb{E}|Z|^r = (\nu-2)^{r/2}\frac{\Gamma\{(r+1)/2\}\Gamma\{(\nu-r)/2\}}{\sqrt{\pi}\,\Gamma(\nu/2)}, \qquad r < \nu. \tag{164}$$

Pointwise $\nu_t > 2$ gives a conditional variance. It does not give a uniform fourth moment or the score moments required by every QMLE theorem. Hansen skew-t, GED, generalized hyperbolic, NIG, Hyperbolic, and Variance Gamma form a collection of native regularity strata rather than one smooth global density chart [22, 24, 5, 6, 26]. Normal is an interior GED point, but it is a boundary limit of Student-t, GH, NIG, and VG. Boundary tests require nonstandard inference.

B.4 CCC, DCC, and conditional copulas

For m series, write the conditional covariance as

$$H_t = D_t R_t D_t, \qquad D_t = \operatorname{diag}(\sqrt{h_{1,t}}, \ldots, \sqrt{h_{m,t}}), \tag{165}$$

where R_t is a positive-definite correlation matrix. A CCC-X model can define

$$R_t = C_m\{C_R b_R(X_t)\}, \tag{166}$$

where $C_m : \mathbb{R}^{m(m-1)/2} \to \mathbb{C}_m^{++}$ is an injective correlation chart. One example is the Archakov–Hansen inverse of

$$\gamma(R) = \operatorname{vecl}(\log R), \tag{167}$$

which maps the interior correlation cone $\mathbb{C}_m^{++}$ bijectively to Euclidean coordinates [2]. Independent Fisher transforms of pairwise correlations do not ensure positive definiteness when $m \geq 3$.

DCC uses an unnormalized positive-definite state:

$$Q_t = c_t\overline{Q}_t + a_t z_{t-1} z_{t-1}' + b_t Q_{t-1}, \tag{168}$$

$$R_t = \operatorname{diag}(Q_t)^{-1/2} Q_t \operatorname{diag}(Q_t)^{-1/2}. \tag{169}$$

A reference-softmax chart can impose

$$(c_t, a_t, b_t) = \frac{(1, e^{\eta_{a,t}}, e^{\eta_{b,t}})}{1 + e^{\eta_{a,t}} + e^{\eta_{b,t}}}. \tag{170}$$

If $\overline{Q}_t$ and Q_0 are positive definite, equations (168) and (170) preserve positive definiteness.

They do not prove strict stationarity of the generative DCC process. DCC may require a nonlinear Markov-chain argument rather than global Lipschitz contraction [20]. Aielli shows that conventional DCC targeting can be inconsistent and proposes corrected DCC [1].

A conditional copula factors the joint density as

$$f_t(y_1, \dots, y_m) = c_{\psi_t}(u_{1,t}, \dots, u_{m,t}) \prod_{i=1}^{m} f_{i,t}(y_i), \qquad u_{i,t} = F_{i,t}(y_i). \tag{171}$$

Here $f_{i,t}$ and $F_{i,t}$ are the i-th marginal density and CDF, and c_{ψ_t} is the copula density. The parameter ψ_t may follow equation (4) or its own observation-driven recursion. Patton gives the conditional-copula validity conditions [29].

C Two-step, boundary, and nonsmooth inference

C.1 Generated inputs and DCC

Let $\gamma_\star$ govern a finite-dimensional first-stage input and suppose

$$\sqrt{n_{\text{eff}}}(\widehat{\gamma} - \gamma_\star) = \frac{1}{\sqrt{n_{\text{eff}}}} \sum_{t \in \mathcal{I}_n} \varphi_{\gamma,t} + o_p(1). \tag{172}$$

Let $g_{c,t}(c, \gamma)$ be the second-stage score and define

$$A_{cc} = \mathbb{E} \frac{\partial g_{c,t}}{\partial c'}, \qquad A_{c\gamma} = \frac{\partial}{\partial \gamma'} \mathbb{E} g_{c,t}(c_\star, \gamma) \Big|_{\gamma = \gamma_\star}. \tag{173}$$

The derivative $A_{c\gamma}$ is total: it includes every effect of the generated input propagated through the recursive filter. Under a joint central limit theorem and the ordinary smooth two-step conditions,

$$\sqrt{n_{\text{eff}}}(\widehat{c} - c_\star) = -A_{cc}^{-1} \frac{1}{\sqrt{n_{\text{eff}}}} \sum_{t \in \mathcal{I}_n} \{ g_{c,t} + A_{c\gamma} \varphi_{\gamma,t} \} + o_p(1). \tag{174}$$

Equation (174) covers root-n, asymptotically linear generated residuals, estimated basis normalizations, and finite-dimensional first-stage parameters. It does not cover a slower or nonregular first stage without orthogonality or a separate functional theorem.

Current DCC estimation commonly fits marginal parameters α first and correlation parameters β second. For this display, take $\mathcal{I}_n = \{1, \dots, n\}$, so $n_{\text{eff}} = n$. Let the estimating equations be

$$\sum_{t=1}^{n} g_{1,t}(\alpha) = 0, \qquad \sum_{t=1}^{n} g_{2,t}(\beta, \widehat{\alpha}) = 0. \tag{175}$$

Define the population Jacobian blocks by

$$A_{ij} = \mathbb{E} \left\{ \frac{\partial g_{i,t}}{\partial \theta_j'} \right\}, \qquad (\theta_1, \theta_2) = (\alpha, \beta). \tag{176}$$

The second-stage influence expansion is

$$\sqrt{n}(\widehat{\beta} - \beta_0) = -A_{22}^{-1} \frac{1}{\sqrt{n}} \sum_{t=1}^{n} \left\{ g_{2,t} - A_{21} A_{11}^{-1} g_{1,t} \right\} + o_p(1). \tag{177}$$

Equation (177) is the Murphy–Topel correction in stacked estimating-equation form [27]. Engle and Sheppard make the generated-residual issue explicit for DCC [18]. If the second-stage filter uses fitted standardized residuals recursively, their full state sensitivity belongs in A_{21}. A covariance correction cannot repair an inconsistent targeting equation.

C.2 Nonregular restrictions

Ordinary Wald, score, and likelihood-ratio limits require an interior point and identified nuisance parameters. Boundary or unidentified cases include

$$\begin{aligned} \tau = 1/\nu = 0 \quad & \text{for a Normal closure limit of Student-}t, \\ a = 0 \quad & \text{for CCC versus DCC,} \\ \gamma = 0 \quad & \text{for no GJR asymmetry,} \\ \alpha = 0 \quad & \text{for a zero ARCH loading.} \end{aligned} \tag{178}$$

The GJR restriction $\gamma = 0$ is regular if signed γ is locally admissible, $\alpha > 0$, and all directions remain identified. It is a boundary restriction if the declared model imposes $\gamma \geq 0$. Let $T_{\mathcal{B}}(c_\star)$ denote the cone of feasible local parameter directions at $c_\star$. When a restriction is identified but lies on a regular boundary, the local estimator obeys

$$\sqrt{n_{\text{eff}}}(\widehat{c}_n - c_\star) \Longrightarrow \operatorname*{arg\,min}_{u \in T_{\mathcal{B}}(c_\star)} \left\{ u'Z + \frac{1}{2} u'Au \right\}, \qquad Z \sim N(0, B). \tag{179}$$

Equation (179) is not an ordinary Gaussian limit [41, 32]. If a nuisance parameter disappears under the null, as DCC persistence can when its dynamic loading is zero, a finite-dimensional cone result is insufficient. Score-driven constancy tests then use a simulated nonstandard process limit [43].

Nonsmooth points require separate request semantics. The examples include

$$e^2 1(e < 0) \quad \text{in GJR,} \qquad |e|^\delta \quad \text{in APARCH,} \qquad |e|^\nu \quad \text{in GED.} \tag{180}$$

Theoretical domains can contain points where a score or Hessian request is undefined. A probability-zero argument may justify asymptotic differentiability under a continuous law. It does not justify returning an exact finite-sample Hessian when a rounded residual equals zero. The derivative request must fail at the nondifferentiable point or use a separately declared nonsmooth estimator.

D Primary-source map

Table 3 lists the role of every main literature strand used in this note. The references are primary papers or author manuscripts. No research note, graph note, or software documentation is used as evidence.

Table 3: Primary sources and the claims for which they are used

Question	Main sources	Role in this note
Varying functions	Hastie and Tibshirani [23]; Chen and Tsay [10]	Basis representation and functional coefficients inside autoregression

Question	Main sources	Role in this note
General causal QMLE	Bardet and Wintenberger [3]; Diop and Kengne [15]	Fixed-parameter general causal models and exogenous-covariate extensions
Smooth time variation	Bardet, Doukhan, and Wintenberger [4]; Dahlhaus, Richter, and Wu [12]; Gao et al. [21]; Karmakar et al. [42]	Locally stationary approximation, local contrast estimation, multivariate examples, and simultaneous bands
Random state existence	Bougerol [9]; Diaconis and Freedman [13]; Doukhan et al. [17]	Random contraction, backward iterations, and random environments
Block and switching stability	Bougerol [9]; Rota and Strang [46]; Straumann and Mikosch [34]	Product-state contraction, joint spectral radius, and function-valued candidate filters
Observation-driven likelihoods	Straumann and Mikosch [34]; Douc et al. [16]; Blasques et al. [37, 38]	Filter forgetting, feasible invertibility, likelihood consistency, derivative filters, and score-driven models
Alternative recursive criteria	Creal et al. [40]; Livieri and Palmari [44]	Moment-driven and proper-score-driven filters; boundaries of the candidate-independent path layer
ARMA paths	Karanasos et al. [25]	Green-function solution and time-varying ARMA summability
Volatility states	Bougerol and Picard [8]; Nelson [28]; Ding et al. [14]; Wintenberger [36]	GARCH stationarity, EGARCH, APARCH, and observed-filter invertibility
Conditional density	Hansen [22]; Rigby and Stasinopoulos [30]; Creal et al. [11]	Predictable density shapes and alternative multi-parameter regressions
Correlation and copulas	Engle [19]; Aielli [1]; Archakov and Hansen [2]; Patton [29]	DCC recursion, corrected targeting, correlation charts, and conditional copulas
Inference	White [35]; Bollerslev and Wooldridge [7]; Rothenberg [31]; Geyer [41]; Self and Liang [32]; Newey and West [45]	Misspecification, QMLE, identification, constraints, nonregular likelihoods, and feasible long-run covariance
Growing sieves	Chen, Liao, and Sun [39]	Existing sieve M-inference and the additional recursive-stability work that a growing path basis would require
Two-step estimation	Murphy and Topel [27]; Engle and Sheppard [18]	Generated-regressor covariance and DCC two-stage estimation

E Certificate to require from each model component

A model component should provide the following mathematical objects before it claims full varying-coefficient support.

1. A state space, metric, native state coordinate, and observation map.
2. A parameter block domain, a regular embedding, and its Jacobian and Hessians.
3. A nonnegative cross-block Lipschitz matrix, anchor envelope, and product-contraction certificate.
4. A timing statement for every modifier and every generated input.
5. A data-generating existence condition for stationary, deterministic, or locally stationary paths.
6. A separate candidate-filter invertibility condition over the optimizer domain.
7. A density normalization and exact moment-existence predicate.
8. First- and second-state sensitivity recursions, forcing moments, and seed-forgetting bounds, including cross blocks.
9. A global or local identification statement for the observable law.
10. A score-limit route: martingale, weak dependence, triangular array, or two-step estimating equations.
11. Boundary and nonsmooth cases with request-specific inference rules.
12. Forecast semantics for future modifier paths.
13. An exact fixed-model nesting test under equation (14).

This certificate is the substantive content behind the common theorem. Once a component supplies it, the basis layer, likelihood assembly, path derivatives, fixed-model restriction, and sandwich inference can be reused.

References

[1] G. P. Aielli. Dynamic conditional correlation: On properties and estimation. *Journal of Business & Economic Statistics*, 31(3):282–299, 2013. https://doi.org/10.1080/07350015.2013.771027.

[2] I. Archakov and P. R. Hansen. A new parametrization of correlation matrices. *Econometrica*, 89(4):1699–1715, 2021. https://doi.org/10.3982/ECTA16910.

[3] J.-M. Bardet and O. Wintenberger. Asymptotic normality of the quasi-maximum likelihood estimator for multidimensional causal processes. *Annals of Statistics*, 37(5B):2730–2759, 2009. https://doi.org/10.1214/08-AOS674.

[4] J.-M. Bardet, P. Doukhan, and O. Wintenberger. Contrast estimation of time-varying infinite memory processes. *Stochastic Processes and their Applications*, 152:32–85, 2022. https://doi.org/10.1016/j.spa.2022.06.005.

[5] O. E. Barndorff-Nielsen. Exponentially decreasing distributions for the logarithm of particle size. *Proceedings of the Royal Society of London A*, 353(1674):401–419, 1977. https://doi.org/10.1098/rspa.1977.0041.

[6] O. E. Barndorff-Nielsen. Normal inverse Gaussian distributions and stochastic volatility modelling. *Scandinavian Journal of Statistics*, 24(1):1–13, 1997. https://doi.org/10.1111/1467-9469.00045.

[7] T. Bollerslev and J. M. Wooldridge. Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2):143–172, 1992. https://doi.org/10.1080/07474939208800229.

[8] P. Bougerol and N. Picard. Stationarity of GARCH processes and of some nonnegative time series. *Journal of Econometrics*, 52(1–2):115–127, 1992. https://doi.org/10.1016/0304-4076(92)90067-2.

[9] P. Bougerol. Kalman filtering with random coefficients and contractions. *SIAM Journal on Control and Optimization*, 31(4):942–959, 1993. https://doi.org/10.1137/0331041.

[10] R. Chen and R. S. Tsay. Functional-coefficient autoregressive models. *Journal of the American Statistical Association*, 88(421):298–308, 1993. https://doi.org/10.1080/01621459.1993.10594322.

[11] D. Creal, S. J. Koopman, and A. Lucas. Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5):777–795, 2013. https://doi.org/10.1002/jae.1279.

[12] R. Dahlhaus, S. Richter, and W. B. Wu. Towards a general theory for nonlinear locally stationary processes. *Bernoulli*, 25(2):1013–1044, 2019. https://doi.org/10.3150/17-BEJ1011.

[13] P. Diaconis and D. Freedman. Iterated random functions. *SIAM Review*, 41(1):45–76, 1999. https://doi.org/10.1137/S0036144598338446.

[14] Z. Ding, C. W. J. Granger, and R. F. Engle. A long memory property of stock market returns and a new model. *Journal of Empirical Finance*, 1(1):83–106, 1993. https://doi.org/10.1016/0927-5398(93)90006-D.

[15] M. L. Diop and W. Kengne. Inference and model selection in general causal time series with exogenous covariates. *Electronic Journal of Statistics*, 16(1):116–157, 2022. https://doi.org/10.1214/21-EJS1950.

[16] R. Douc, P. Doukhan, and E. Moulines. Ergodicity of observation-driven time series models and consistency of the maximum likelihood estimator. *Stochastic Processes and their Applications*, 123(7):2620–2647, 2013. https://doi.org/10.1016/j.spa.2013.04.010.

[17] P. Doukhan, M. H. Neumann, and L. Truquet. Stationarity and ergodic properties for some observation-driven models in random environments. *Annals of Applied Probability*, 33(6B):5145–5170, 2023. https://doi.org/10.1214/23-AAP1944.

[18] R. F. Engle and K. Sheppard. Theoretical and empirical properties of dynamic conditional correlation multivariate GARCH. NBER Working Paper 8554, 2001. https://doi.org/10.3386/w8554.

[19] R. F. Engle. Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics*, 20(3):339–350, 2002. https://doi.org/10.1198/073500102288618487.

[20] J.-D. Fermanian and H. Malongo. On the stationarity of the dynamic conditional correlation model. *Econometric Theory*, 33(3):636–663, 2017. https://doi.org/10.1017/S0266466616000116.

[21] J. Gao, B. Peng, W. B. Wu, and Y. Yan. Time-varying multivariate causal processes. *Journal of Econometrics*, 240(1):105671, 2024. https://doi.org/10.1016/j.jeconom.2024.105671.

[22] B. E. Hansen. Autoregressive conditional density estimation. *International Economic Review*, 35(3):705–730, 1994. https://doi.org/10.2307/2527081.

[23] T. Hastie and R. Tibshirani. Varying-coefficient models. *Journal of the Royal Statistical Society: Series B*, 55(4):757–796, 1993. https://doi.org/10.1111/j.2517-6161.1993.tb01939.x.

[24] E. Jondeau and M. Rockinger. Conditional volatility, skewness, and kurtosis: Existence, persistence, and comovements. *Journal of Economic Dynamics and Control*, 27(10):1699–1737, 2003. https://doi.org/10.1016/S0165-1889(02)00079-9.

[25] M. Karanasos, A. Paraskevopoulos, T. Magdalinos, and A. Canepa. A unified theory for ARMA models with varying coefficients: One solution fits all. *Econometric Theory*, 42:192–245, 2026. https://doi.org/10.1017/S0266466624000306.

[26] D. B. Madan, P. P. Carr, and E. C. Chang. The variance gamma process and option pricing. *European Finance Review*, 2(1):79–105, 1998. https://engineering.nyu.edu/sites/default/files/2018-09/CarrEuropeanFinReview1998.pdf.

[27] K. M. Murphy and R. H. Topel. Estimation and inference in two-step econometric models. *Journal of Business & Economic Statistics*, 20(1):88–97, 2002. https://doi.org/10.1198/073500102753410417.

[28] D. B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991. https://doi.org/10.2307/2938260.

[29] A. J. Patton. Modelling asymmetric exchange rate dependence. *International Economic Review*, 47(2):527–556, 2006. https://doi.org/10.1111/j.1468-2354.2006.00387.x.

[30] R. A. Rigby and D. M. Stasinopoulos. Generalized additive models for location, scale and shape. *Journal of the Royal Statistical Society: Series C*, 54(3):507–554, 2005. https://doi.org/10.1111/j.1467-9876.2005.00510.x.

[31] T. J. Rothenberg. Identification in parametric models. *Econometrica*, 39(3):577–591, 1971. https://doi.org/10.2307/1913267.

[32] S. G. Self and K.-Y. Liang. Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610, 1987. https://doi.org/10.1080/01621459.1987.10478472.

[33] A. Sorokin. Non-invertibility criteria in some heteroscedastic models. Preprint, 2011. https://arxiv.org/abs/1104.3318.

[34] D. Straumann and T. Mikosch. Quasi-maximum-likelihood estimation in conditionally heteroscedastic time series: A stochastic recurrence equations approach. *Annals of Statistics*, 34(5):2449–2495, 2006. https://doi.org/10.1214/009053606000000803.

[35] H. White. Maximum likelihood estimation of misspecified models. *Econometrica*, 50(1):1–25, 1982. https://doi.org/10.2307/1912526.

[36] O. Wintenberger. Continuous invertibility and stable QML estimation of the EGARCH(1,1) model. *Scandinavian Journal of Statistics*, 40(4):846–867, 2013. https://doi.org/10.1111/sjos.12038.

[37] F. Blasques, P. Gorgi, S. J. Koopman, and O. Wintenberger. Feasible invertibility conditions and maximum likelihood estimation for observation-driven models. *Electronic Journal of Statistics*, 12(1):1019–1052, 2018. https://doi.org/10.1214/18-EJS1416.

[38] F. Blasques, J. van Brummelen, S. J. Koopman, and A. Lucas. Maximum likelihood estimation for score-driven models. *Journal of Econometrics*, 227(2):325–346, 2022. https://doi.org/10.1016/j.jeconom.2021.06.003.

[39] X. Chen, Z. Liao, and Y. Sun. Sieve inference on possibly misspecified semi-nonparametric time series models. *Journal of Econometrics*, 178(Part 3):639–658, 2014. https://doi.org/10.1016/j.jeconom.2013.10.002.

[40] D. Creal, S. J. Koopman, A. Lucas, and M. Zamojski. Observation-driven filtering of time-varying parameters using moment conditions. *Journal of Econometrics*, 238(2):105635, 2024. https://doi.org/10.1016/j.jeconom.2023.105635.

[41] C. J. Geyer. On the asymptotics of constrained M-estimation. *Annals of Statistics*, 22(4):1993–2010, 1994. https://doi.org/10.1214/aos/1176325768.

[42] S. Karmakar, S. Richter, and W. B. Wu. Simultaneous inference for time-varying models. *Journal of Econometrics*, 227(2):408–428, 2022. https://doi.org/10.1016/j.jeconom.2021.03.002.

[43] Y. Lin and A. Lucas. Testing for the absence of score-driven parameter dynamics. Tinbergen Institute Discussion Paper 2025-063/III, 2025. https://papers.tinbergen.nl/25063.pdf.

[44] G. Livieri and G. Palmari. Proper-score observation-driven filters: Local geometry, estimation, and continuous-time limits. arXiv:2608.02828, 2026. https://arxiv.org/abs/2608.02828.

[45] W. K. Newey and K. D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987. https://doi.org/10.2307/1913610.

[46] G.-C. Rota and W. G. Strang. A note on the joint spectral radius. *Indagationes Mathematicae (Proceedings)*, 63:379–381, 1960. https://doi.org/10.1016/S1385-7258(60)50046-1.